

Exploring the Potential of Generative AI in Song-Signing

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Abstract

This study explores the potential of generative AI technology to support song-signing creation and proposes a design for an AI-based song-signing authoring tool. We identified the creative process, strategies, challenges, and requirements for an AI-assisted tool through focus group interviews with two song-signing creators. The findings suggest that generative AI can be applied to automate and streamline various aspects of the song-signing creation process, including music analysis, lyrics translation, and choreography generation. Based on these insights, we propose a design of an AI-based song-signing authoring tool incorporating large language models for sign language translation and recommendation, generative dance AI for sign language visualization, and customizable sign language avatars. The development of such tools has the potential to expand the participation of d/Deaf individuals in cultural and artistic activities and contribute to an inclusive cultural and creative ecosystem.

CCS Concepts

• Human-centered computing → Field studies.

Keywords

Song-signing, Generative AI, Large Language Model (LLM), d/Deaf individuals, Cultural Inclusivity

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1 Introduction

Music plays a crucial role in the development of human social and emotional skills, yet it remains a challenging domain for individuals with hearing impairments to access [4, 7–9]. In this context, song-signing, a visually oriented form of music that expresses songs

through sign language, has emerged as a significant musical genre within the d/Deaf¹ culture [3, 17]. Song-signing transcends the mere translation of music into sign language; it visually conveys the musical elements, evolving into a new art form. Furthermore, it serves as a medium for cultural exchange between the non-d/Deaf and d/Deaf communities, fostering the development of diverse musical styles and holding value as a cultural phenomenon [14].

However, the process of creating song-signing content involves several challenges. Translating lyrics from spoken language to sign language presents difficulties due to linguistic differences, and conveying the emotions and rhythms of music through sign language expressions requires skilled techniques and considerable effort [26].

Recent advancements in generative AI technology have shown promising potential in assisting creative processes across various domains [27]. Generative AI learns from existing data to generate new content and has been successfully utilized in fields such as image generation, music composition, storytelling, and dance choreography [16, 19, 23]. By leveraging AI, it may be possible to automate and streamline the song-signing creation process, particularly the generation of dance through music analysis, thereby reducing the burden on creators and enabling a wider range of individuals to participate in this art form.

The goal of this study is to design a generative AI-based song-signing authoring tool system based on the experiences and requirements of creators. To achieve this, we conducted focus group interviews (FGIs) with two sign language interpreters who perform song-signing, eliciting the difficulties encountered in the creative process and the requirements for an AI-based co-creative tool. This paper introduces the value and creative process of song-signing and proposes a design direction for an authoring tool system that leverages generative AI technology. By enhancing the efficiency of song-signing content creation, we aim to ultimately contribute to expanding musical enjoyment opportunities for individuals with hearing impairments.

2 Related work

2.1 Song-Signing Music

Song-signing is a musical content that involves translating and expressing music through sign language [3, 17, 26]. It goes beyond the mere translation of song lyrics into sign language; it visually



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¹The lower case “deaf” refers to the medical condition of deafness with no residual hearing, and the upper case “Deaf” means that they identify as part of the Deaf community.

conveys the melody, rhythm, and emotions of the music through movements and facial expressions, providing a unique musical experience [22]. Song-signing has been enjoyed within the d/Deaf culture for a long time, and recently, with the development of video-sharing platforms like YouTube, it has gained interest not only from the d/Deaf community but also from the non-d/Deaf group through active exchanges [25]. Song-signing transcends the notion of sound as a deficit and instead embraces a perspective that recognizes individuals with different sensory experiences as valued users of music. This approach contributes to the development of diverse musical styles and positions song-signing as an important cultural phenomenon [18].

The creative process of song-signing involves several challenges, such as translating lyrics from spoken language to sign language, conveying the emotions and rhythms of music through sign language expressions, and the scarcity of specialized performers [18, 26]. These challenges highlight the need for innovative solutions to support the creation and dissemination of song-signing content.

2.2 Generative AI for Accessibility

The advancements in generative AI technology have opened new possibilities for people with limited artistic expertise to engage in creative activities [11, 24, 27]. By leveraging artistic concepts and simple sketches or prompt inputs, individuals can collaborate with AI to generate creative content, making artistic expression more accessible to groups that previously faced barriers, such as the elderly and people with disabilities [12, 15]. Researchers have begun to actively explore the use of AI-based tools to support artistic creation for individuals with disabilities. For example, sign language recognition and generation models for those with hearing impairments [10], and AI-based digital art tools for individuals with cognitive disabilities [12]. These initiatives demonstrate the potential of generative AI technology in empowering people with disabilities to express themselves creatively and participate in artistic endeavors.

In the context of song-signing, generative AI technology could play a crucial role in automating and streamlining the content creation process, thereby reducing the reliance on specialized creators, and enabling a wider range of individuals to engage in this unique art form. By leveraging AI-based tools to assist with tasks such as lyric translation, dance generation, and the incorporation of musical elements, the challenges associated with song-signing creation could be mitigated, leading to increased accessibility and diversity within this cultural phenomenon.

3 Focus Group Interview

3.1 Participants

The focus group interview (FGI) was conducted with two experienced song-signing creators who have been actively involved in producing and performing song-signing content. Both participants are professional sign language interpreters with extensive experience in the field. Participant A is a 40-year-old male with 10 years of experience as a sign language interpreter and has performed song-signing in more than 6 events. Participant B is a 32-year-old male with 8 years of experience as a sign language interpreter and has also performed song-signing in over 5 events.

3.2 Process

The FGI was conducted using a semi-structured interview approach to gather insights into the song-signing creation process and explore the potential for integrating generative AI technologies. The interview lasted approximately 2 hours and was guided by a predefined set of questions. The primary goals of the FGI were twofold: 1) to gain a comprehensive understanding of the song-signing creation process for performances and 2) to assess the feasibility and potential applications of generative AI technologies within this domain. The interview covered various aspects of song-signing creation, including the creative process, methods and strategies employed, difficulties faced by creators, and the perceived benefits and concerns regarding the use of AI co-creative tools by introducing technologies and services for using AI co-creation tools. The recorded video was transcribed and coded independently by two researchers who participated in the interview using a thematic analysis approach [6].

4 Findings

4.1 Song-Signing Creative Process

The participants divided the song-signing creative process into three main stages: music analysis, lyrics translation, and music expression (Figure 1). In the music analysis stage, they identified the genre, tempo, rhythm, and the meaning and emotions of the lyrics. Lyrics translation involves converting the original lyrics into sign language. They then carefully considered the grammatical differences between spoken language and sign language, as well as the characteristics of visual expression, employing both literal and interpretive translation as appropriate. Finally, through repetitive practice, they expressed and learned the sign language choreography, and refined their emotional expressions to match the music. Both participants reported listening to a song an average of 300 times or more to complete the creative process.

4.2 Strategies for Creating Song-Signing

4.2.1 Sign Language Lyrics Translation Strategies. The participants revealed several strategies they use when creating sign language lyrics. First, they semantically translate the meaning of the song lyrics into sign language. Second, they visually imagine the scenes expressed in the lyrics and convey them through sign language. Third, they represent the mood and emotions of the music through sign language gestures. In this process, the linguistic characteristic of sign language, spatial utilization, is carefully considered. Unlike linear spoken language, sign language is a three-dimensional language that can convey spatial information that spoken language cannot express. In particular, the direction and intensity of gestures and facial expression are important.

4.2.2 Expressing Musical Elements. The participants stated that they strive to express musical elements such as tempo, rhythm, and mood through sign language gestures and non-manual signals. They particularly emphasized the challenge of visualizing the instrumental parts of a song where there are no lyrics. To address this, they utilize performances such as moving their bodies or facial expressions in sync with the music.

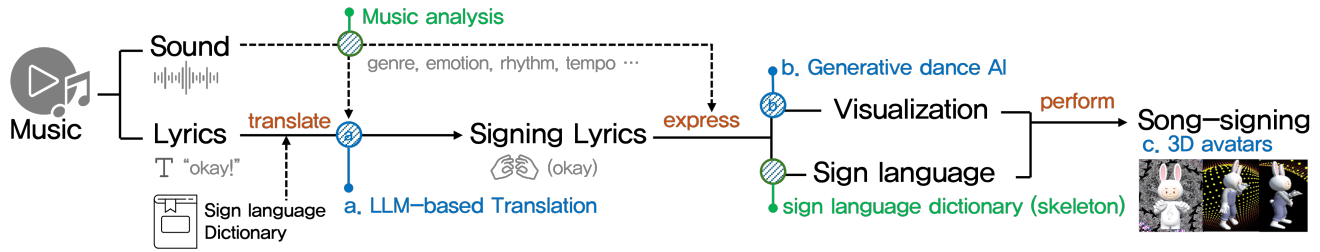


Figure 1: Overall architecture of AI-assisted song-signing Creation. The process illustrates the potential applications of AI technology in the song-signing process based on the creative strategies employed by the participants.

4.3 Challenges of Creating Song-Signing

4.3.1 Difficulty in Lyrics Translation. The participants highlighted the challenges involved in translating spoken language lyrics into sign language. They emphasized that the meaning and nuances of the original lyrics can often be lost or altered during the translation process. Symbolic concepts and metaphorical expressions, which are common in song lyrics, are particularly difficult to convey in a visual language like sign language. The participants also noted that finding the appropriate sign language vocabulary to accurately represent the lyrics can be a time-consuming and arduous task, as not all words or phrases have a direct equivalent in sign language. This challenge can lead to a reduction or distortion of the original meaning when the lyrics are translated into sign language.

4.3.2 Limitations in Expression. The participants acknowledged that creating a compelling song-signing performance requires a combination of sign language proficiency and musical expression abilities. However, they pointed out that individuals vary significantly in their expressive capabilities, largely due to differences in their inherent musicality and creativity. Some individuals may have a natural aptitude for conveying emotions and the mood of the music through their sign language performance, while others may struggle to do so. The participants specifically highlighted the difficulty in determining how to visually express the non-lyrical parts of a song, as these sections rely heavily on the performer's ability to interpret and convey the music's essence through their movements and facial expressions.

4.3.3 Time and Effort Required for Creation. The participants emphasized that creating a polished song-signing performance is a highly time-consuming and labor-intensive process. They explained that completing a single song can require hundreds of practice sessions, starting from the initial translation of the lyrics into sign language, to the development of the sign language choreography, and finally, the repetitive rehearsals to perfect the performance. The participants stressed that refining the sign language expressions to accurately capture the mood and emotions of the music, while simultaneously synchronizing the signing with the timing of the song, is a challenging and demanding task. This process requires a significant investment of time and effort from the song-signing creators and performers.

4.3.4 Lack of Song-signing creators and performers. Although interest in song-signing is increasing with the growing demand for

barrier-free performances, there is a shortage of performers skilled in song-signing expression. As it requires specialized knowledge in sign language and various skills to express music, it is challenging for both non-d/Deaf and d/Deaf individuals to easily create song-signing content. Both participants emphasized the need to lower the entry barrier to the creative process and support the participation of individuals from diverse backgrounds to enable the wider spread of song-signing. They stressed that this would allow both non-d/Deaf individuals lacking sign language expertise and d/Deaf individuals lacking musical skills to create song-signing content.

4.4 Possibilities of Generative AI in Creating Song-Signing

In this study, we introduced existing Generative AI technologies and services and sought participants' opinions on how generative AI could be utilized in the song-signing creative process. The participants anticipated that generative AI could reduce the burden on creators by offering, 1) sign language translation and recommendation functions during the lyrics translation process and 2) choreography generation for visual expression of musical elements based on music analysis. Additionally, they suggested that avatar could be an alternative for creating song-signing content. However, they emphasized that when introducing technology, the intentions and interpretations of the creators should be reflected, and caution should be taken to ensure that excessive automation does not limit creativity [20]. Based on the participants' opinions, we propose the following AI-based song-signing authoring tools:

4.4.1 LLM-based Sign Language Translation and Recommendation. We propose a function that utilizes Large Language Models (LLMs) to automatically translate or recommend song lyrics into sign language lyrics. LLM models can understand the meaning of lyrics through sign language dictionaries and generate appropriate sign language expressions. In this process, it is necessary to optimize the translation results by considering the differences in expression length between spoken lyrics and sign language lyrics. Furthermore, it is desirable to select the most suitable sign language gestures among expressions with the same meaning that best match the characteristics of the music. For example, in (Figure 2), when expressing the word 'umbrella', sign language A with a small gesture is appropriate for static ballads, while sign language B is more appropriate for dynamic music such as rock bands. Additionally, it would be necessary to provide a function that allows creators to

select and edit the translated sign language lyrics to reflect their intentions.

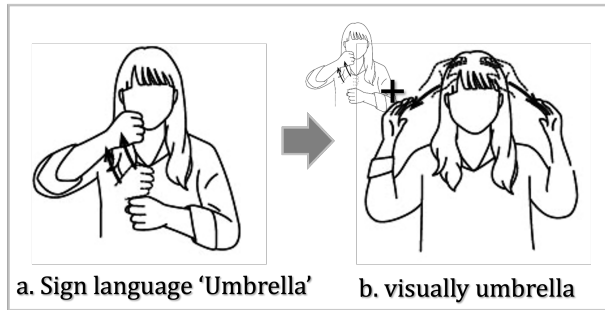


Figure 2: Example of sign language selection considering the visual representation of music

4.4.2 Sign Language Visualization through Generative Dance AI. We propose an artificial intelligence model that analyzes the mood, rhythm, and emotions of a song and automatically generates choreography accordingly [23]. The translated sign language lyrics are mapped to the upper body of a virtual avatar, while the generated dance choreography is mapped to the lower body. However, this distinct mapping may reduce the visibility of the sign language lyrics. Therefore, the dance generation model should not only analyze musical elements but also consider the unique visual expression methods of sign language, such as staging, when composing the choreography. Both participants agreed that to prevent the limitation of creativity due to excessive automation and to reflect the creators' intentions, a parameter adjustment function should be provided to allow the selection and modification of the generated choreography. In particular, since the meaning of sign language varies depending on the direction, parameters to adjust the direction of movements or options to control the size and intensity of movements that express the mood of the music can be considered.

4.4.3 Utilization of Sign Language Avatars. 3D avatar technology has been recognized as a suitable tool for realizing the three dimensional expression of sign language. One of its key advantages is the ability to effectively reproduce spatial utilization, which is a core feature of sign language communication. However, for the avatar to go beyond simply mimicking sign language and convey the mood and feel of the music, it is necessary to support a customization function that allows the avatar's design to be tailored according to the characteristics of each song. Moreover, to effectively communicate not only the meaning of the sign language lyrics but also the musical atmosphere, a facial expression design function for the avatar is required. Participants in the study unanimously agreed that such an avatar could greatly contribute to the implementation of barrier-free performances, even in the absence of a live sign language performer.

4.4.4 Tailored AI Assistance for Diverse Users. As both d/Deaf and non-d/Deaf individuals face challenges in creating song-signing, there is a need for AI architectures designed to meet the specific requirements of these two groups. For non-d/Deaf users, enhanced

sign language translation is necessary, including real-time feedback on sign accuracy and cultural nuances. Additionally, visual guides for proper hand positioning and movement in sign language are crucial to support their learning process. d/Deaf users, on the other hand, require designs that focus more on sound accessibility. AI-powered visual representations of musical elements such as rhythm, tempo, and melody are essential. Furthermore, haptic feedback systems to convey musical timing and beat can greatly enhance their understanding and expression of musical aspects.

These tailored AI assistance approaches aim to address the unique challenges faced by each group, potentially lowering the barrier to entry and expanding the pool of song-signing creators and performers from both communities.

5 Discussion

From a socio-cultural perspective, generative AI-based song-signing authoring tools can contribute to expanding the participation of d/Deaf individuals in cultural and artistic activities and enhancing their creative expression abilities. Due to the lack of specialized skills and resources required for song-signing creation, the activities of d/Deaf artists have been limited. However, AI-powered creation tools can lower these entry barriers and provide opportunities for more creators to participate in song-signing content production [12]. This can lead to the expansion of the base and diversity of cultural and artistic activities for people with disabilities. Furthermore, by facilitating collaboration between d/Deaf and non-d/Deaf artists through AI technology, it can serve as an opportunity to broaden the scope of communication and understanding between different cultures. Song-signing content created through technology can touch a wider audience and help improve awareness of disability culture. Therefore, the development of generative AI-based song-signing creation tools can play an important role in fostering an inclusive cultural and artistic ecosystem.

However, it is necessary to be cautious that the development and utilization of technology do not lead to the distortion or marginalization of the creativity of specific groups [13, 24]. To ensure that technology is used to support the diversity of creators and strengthen their capabilities, efforts should be made to actively reflect the voices of stakeholders and establish ethical guidelines during the development process. Ultimately, generative AI technology should be positioned as a tool that supports and expands the creative expression of human creators while respecting their autonomy and diversity [11, 13].

While our proposed architecture follows the three-stage process identified in the focus group interviews, it's important to acknowledge that modern multimodal machine learning models often employ end-to-end approaches [5]. This presents an opportunity to bridge the gap between cutting-edge ML techniques and the practical system needs emphasized in HCI studies [2]. Future research could explore hybrid approaches that combine the efficiency of end-to-end models with the interpretability and user-centric design principles valued in HCI [1]. For instance, a single multimodal model could potentially handle music analysis, lyrics translation, and expression generation simultaneously, while still providing interpretable outputs at each stage to support user interaction and customization. Such an approach could potentially enhance the

efficiency and effectiveness of song-signing creation tools, while maintaining the user-centered focus crucial in accessibility applications [21].

6 Conclusion

In conclusion, this study explores the potential of generative AI in supporting song-signing creation and proposes a design for an AI-based authoring tool. Through interviews with creators, we identified the creative process, challenges, and the role of AI in assisting them. The findings suggest that generative AI can be applied to various aspects of song-signing creation, including music analysis, sign language translation, choreography generation, and AI avatars. We propose an AI-based authoring tool incorporating these features, which has the potential to innovate the creative process, enhance efficiency, and expand the participation of d/Deaf individuals in artistic activities, contributing to a more inclusive ecosystem. However, future research should involve a broader range of stakeholders further to refine the design of AI-based song-signing authoring tools and address potential ethical concerns, ensuring that the development and utilization of AI technology respect the autonomy and diversity of human creators.

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